

Comparison of metaheuristic optimization algorithms for RSS-based 3-D visible light positioning systems

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Abstract—In this work we present a comparison between two classical metaheuristic optimization algorithms to solve two different received signal strength (RSS) based fitness function formulations for 3-D visible light positioning systems. For this purpose, simulations have been carried out to analyze the performance of genetic algorithm (GA) and particle swarm optimization (PSO) methods for mobile node's position determination. Based on simulation results, GA overcomes PSO. GA shows to be more robust than PSO and it is capable to provide feasible optimal solutions when no map information is added as constraints to the optimization problem. When map information is used as lower and upper bound constraints, GA and PSO methods increase their accuracy and precision significantly.

Index Terms—Visible light positioning, optimization, visible light communications

I. INTRODUCTION

Localization based services have become part of our lives in the last decades. In particular, the global positioning system (GPS) has been a fundamental technological platform to provide various of these services in outdoor scenarios. Nevertheless, despite the qualities exhibited in outdoor scenarios, GPS based systems have been unable to overcome different challenges when used for indoor or metropolitan environments. In order to provide indoor positioning service, different alternatives have been proposed through the years. Most of nowadays proposals are based on radio frequency (RF) technology. The usage of technologies such as RFID, Bluetooth, Wi-Fi and Zigbee has been proposed in literature [1], [2]. These systems are severely affected by multipath reflections when used in indoor environments. Due to this, they provide positioning with an accuracy of tens of centimetres to few meters.

Visible light positioning (VLP) has gained attention of researchers. VLP systems are less susceptible to multipath. Therefore, they provide positioning with higher accuracy in environments compared to RF-based solutions. Another advantage of these systems is that they use already deployed illumination/communication infrastructure [3] to provide local-

ization services. Moreover, they can be used in environments where RF systems are limited or forbidden such as hospitals and airplanes.

VLP systems can be classified in two different groups, i.e., image sensor (IS) based positioning systems and photo detector (PD) based positioning systems [4]. This classification is done based on the signal detection hardware. Image sensors provide LED position information in a more direct fashion than PD counterpart. Furthermore, no multiplexing technique is required since multiple light sources can be directly separated from the image [3]. Moreover, nowadays off-the-shelf devices have incorporated cameras. The main disadvantage of IS-based VLC networks is that they can provide a very limited data rate. In addition to this, it demands a high computational effort to process the image stream.

On the other hand, VLC systems based on PD provide very high data rate communication [4], [5]. PDs are easy to install and inexpensive. Moreover, the received signal can be treated at a lower computational cost than the required by IS-based systems. Contrary to IS-based positioning, the usage of multiplexing techniques is required in PD-based VLP systems. Depending on the multiplexing technique, the required hardware can increase in complexity and affect the performance of the positioning method [6], [7]. In [8], [9] the performance of time division multiplexing (TDM), frequency division multiplexing (FDM) and code division multiplexing (CDM) techniques for VLP were investigated. In particular, FDM systems obtain better performance in terms of accuracy when compared with TDM and CDM.

Localization methods can be classified in two types, i.e., range based and range free [10]. Range based methods require an estimation of the distance between transmitter and receiver. This estimation is done by measuring the properties of the received signal. On the other hand, range free algorithms use only connectivity information to perform localization. Due to this, no extra hardware is required to measure the signal characteristics. Nevertheless, range free algorithms suffer from

low accuracy.

Some of the most used received signal characteristics for ranging determination in range based methods are: received signal strength (RSS), time of arrival (TOA), time difference of arrival (TDOA) and angle of arrival (AOA). A few number of surveys can be found in literature where the performance of positioning methods based on these signal characteristics has been analysed in terms of accuracy [11]. Compared to other signal characteristics, RSS values can be easily obtained using low cost and off-the-shelf devices. In addition to this, RSS positioning is easier than TDOA and AOA methods and achieves high accuracy. Due to this, RSS is the preferred technology for implementing IPS. There are many localization methods that use RSS to perform positioning, i.e., trilateration, fingerprinting and proximity. In order to perform trilateration, a propagation model is required. In this model, the intensity is reduced as the distance between the beacon node and the mobile node [11]. Received intensity can be affected by shadowing, multipath reflections and others. This, will lead to a detrimental effect on the range estimation and, as a consequence, on the algorithm accuracy [12]. Previous works of different authors have pointed out the need of strong global optimization heuristic methods for 3-D position estimation. Although few authors have proposed their own particular methods for 3-D positioning, these methods increase the hardware complexity of the network or the computational cost of position determination. Another approach used in 3-D positioning system based on visible light communications is the usage of heuristic optimization methods in order to find the optimal solution of a received signal strength (RSS) based non-linear least-square (NLLS) problem for positioning.

In [13] the authors use differential evolution algorithm to solve a positioning NLLS optimization problem. Similarly, in [14], [15] the authors address the problem using a PSO based method. In [16] a modified genetic algorithm is used to determine the mobile node's position. None of these authors have compared their proposed methods with other metaheuristic optimization algorithms. This is an important step when using metaheuristic optimization algorithms since their performance depends on the optimization problem (problem dependent methods).

In this paper we analyze the impact of optimization problem formulation in the performance of the optimization method. For doing this, two different Non-Linear Least Squares (NLLS) minimization problems are proposed for position determination. Classical genetic algorithm (GA) and particle swarm optimization (PSO) are compared based on their performance in terms of accuracy and precision. The work is organized as follows. In Section II the proposed system for simulation and demonstration are presented. Optimization methods used for comparison are shown in Section III. Results are presented and discussed in Section IV. Finally, the main conclusions of the study and future work will be presented in Section V.

II. SYSTEM DESCRIPTION AND PROPOSED METHOD

A. System model

We assume an homogeneously distributed lighting deployment of 4 LEDs lights as shown in Fig. 1. Each light is separated by the same distance $D_x = D_y$.

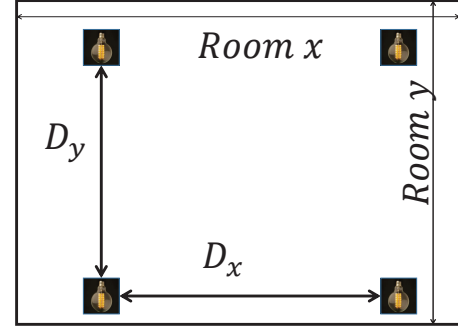


Fig. 1. LED deployment for the proposed scenario

The total received power can be expressed by the contribution of line-of-sight (LOS), non-LOS and noise [17] as detailed in Eq. 1

$$P_{ij} = R_{PD_j} P_i H_{ij}^{LOS} + R_{PD_j} P_i H_{ij}^{non-LOS} + N \quad (1)$$

where P_{ij} is the received power at mobile node j from LED i , R_{PD_j} , P_i and H_{ij}^{LOS} are the responsivity of the photo detector, the transmitted power by the i^{th} LED and the LOS channel gain respectively. The non-LOS channel gain is expressed as $H_{ij}^{non-LOS}$. The noise N is additive noise modelled as the sum of thermal noise $\sigma_{thermal}$ and shot noise σ_{shot} [18]. The LOS and non-LOS channel gain components of the VLC link are graphically shown in Fig. 2.

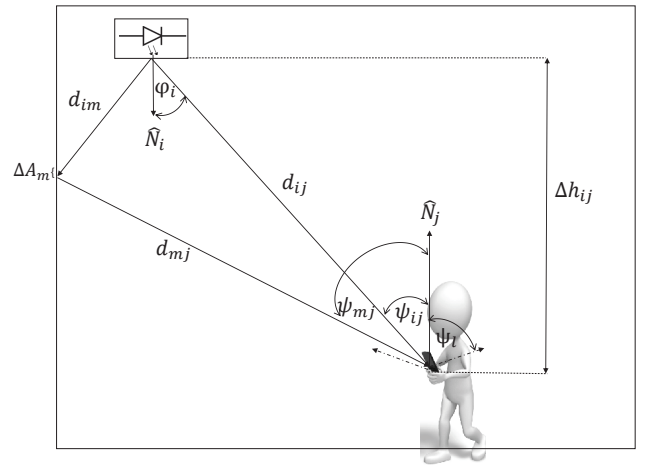


Fig. 2. Visible light communications channel model

The LOS component can be mathematically expressed as [17]

$$H_{ij}^{LOS} = \begin{cases} \frac{(m_l+1)A}{2\pi d_{ij}^2} \cos^{m_l}(\varphi_{ij}) G(\psi_{ij}) \cos(\psi_{ij}) & 0 \leq \psi_{ij} \leq \Psi_l \\ 0 & \text{elsewhere} \end{cases} \quad (2)$$

where m_l is the Lambertian order transmission, A is the effective area of the PD, d_{ij} is used to denote the distance between transmitter and receiver. Finally, φ_{ij} and ψ_{ij} are the angle of irradiance and the angle of incidence of the signal. $G(\psi_{ij}) = T_s(\psi_{ij})g(\psi_{ij})$ is the product of the optical filter gain and the optical concentrator. The field of view (FOV) of the optical receiver Ψ_l plays an important role in VLC communications since this parameter will determine whether the receiver is capable to perceive or not the source based on the incidence angle ψ_{ij} . The FOV value depends in the quality and type of the photodiode. The non-LOS channel gain shown in Fig. 2 can be mathematically expressed as

$$H_{ij}^{non-LOS} = \sum_{m=1}^K H_{im} H_{mj} \delta\left(t - \frac{d_{mi} + d_{mj}}{c}\right) \quad (3)$$

where H_{im} is the channel gain from LED to wall and H_{mj} is the channel gain from wall to mobile node. In particular each gain is obtained as follows

$$H_{im} = \frac{\Delta A_m \rho_m}{d_{im}^2} \cos^{m_l}(\phi_{im}) \cos(\psi_{im}) \quad (4)$$

$$H_{mj} = \frac{(m_l+1)A}{2\pi d_{mj}^2} \cos(\phi_{mj}) \cos(\psi_{mj}) G(\psi_{mj}) \quad (5)$$

B. Proposed methods for RSS-based 3-D positioning

As shown in Eq. 1 the received power is related to the transmitted power, received channel gain and noise. The LOS channel gain H_{ij}^{LOS} can be re-written as

$$H_{ij}^{LOS} = \begin{cases} \frac{C}{d_{ij}^2} \cos^{m_l}(\varphi_{ij}) \cos(\psi_{ij}) & 0 \leq \psi_{ij} \leq \Psi_l \\ 0 & \text{elsewhere} \end{cases} \quad (6)$$

where C is a constant parameter based on characteristics of the VLC system. In order to simplify the analysis, we consider co-planar condition of emitter and receiver. Due to this, irradiance angle φ_{ij} and incidence ψ_{ij} are equal. Both angles can be expressed in terms of the relative position between transmitter and receiver as

$$\varphi_{ij} = \psi_{ij} = \frac{z_i - z_j}{d_{ij}} \quad (7)$$

were z_j and z_i are the height of receiver and transmitter respectively. By substituting Eq. 7 into Eq. 6 we obtain

$$H_{ij}^{LOS} = \begin{cases} C \frac{(z_i - z_j)^{m_l+1}}{(d_{ij})^{m_l+3}} & 0 \leq \psi_{ij} \leq \Psi_l \\ 0 & \text{elsewhere} \end{cases} \quad (8)$$

The distance between transmitter and receiver is also obtained using the relative positions of transmitter and receivers

as $d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}$. In order to determine the position of the mobile node, we minimize the difference between the estimated channel gain and the obtained channel gain as follows

$$\begin{aligned} & \underset{(\tilde{x}_j, \tilde{y}_j, \tilde{z}_j)}{\text{minimize}} && \sum_{i=1}^N \left(\tilde{H}_{ij}^{LOS} - \frac{P_{ij}}{P_i} \right)^2 \\ & \text{subject to} && (\tilde{x}, \tilde{y}, \tilde{z}) \succeq 0 \end{aligned} \quad (9)$$

where $(\tilde{x}_j, \tilde{y}_j, \tilde{z}_j)$ is the estimated position of the mobile node by the optimization algorithm and N is the total number of received sources. The estimated channel gain $\tilde{H}_{ij}^{LOS}$ is obtained by replacing the position guess into Eq. 8. This problem is only constrained in its lower bounds as the position estimates must be greater or equal to zero. In order to measure the impact of including room size (map) information we will compare the results of solving the optimization problem of Eq. 9 with the following lower and upper bounded approach

$$\begin{aligned} & \underset{(\tilde{x}_j, \tilde{y}_j, \tilde{z}_j)}{\text{minimize}} && \sum_{i=1}^N \left(\tilde{H}_{ij}^{LOS} - \frac{P_{ij}}{P_i} \right)^2 \\ & \text{subject to} && (\tilde{x}, \tilde{y}, \tilde{z}) \succeq 0, \\ & && (\tilde{x}, \tilde{y}, \tilde{z}) \preceq (R_x, R_y, R_z) \end{aligned} \quad (10)$$

where R_x , R_y and R_z are the maximum length, width and height of the considered room (map information).

III. METAHEURISTIC GLOBAL SEARCH OPTIMIZATION METHODS

Global optimization is used to find the global minima or maxima. Finding the global minima of a function may not always be done due to several reasons. Frequently, analytical methods are not applicable for most of the optimization problems since they are likely to get stuck in a local optimum. On the other hand, metaheuristic global optimization methods are designed to provide a sufficiently good solution to an optimization problem. In Fig. 3 a graphical representation of the NLLS optimization problem is shown.

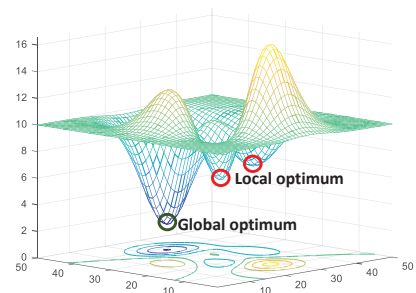


Fig. 3. Graphical representation of the non-linear least-square (NLLS) problem

Metaheuristic optimization methods are capable to find good solutions using less computational effort than other

classical algorithms. Metaheuristics are particularly useful for non-linear and non-convex problems where global optimum solutions can not be found using traditional methods.

In this work we will compare two of the most classical metaheuristic global optimization methods, i.e., genetic algorithms and particle swarm optimization.

1) *Genetic Algorithm (GA)*: Genetic Algorithms are inspired in natural selection and the survival of the fittest. GA is a population-based optimization method and work by randomly creating a first population of individuals (solutions). A possible solution of an optimization problem is encoded into the form of an individual “chromosome”, which is a solution for the optimization problem. The chromosome is composed by a set of elements called genes. If an optimization problem has V variables to be found, then the chromosome is encoded into a V -element array

$$\text{Chromosome} = [x_1, x_2, \dots, x_V] \quad (11)$$

where x_i represents a variable on the optimization problem.

The fitness of each individual is calculated according to an objective function. Best individuals are selected based on their “fitness” to pass their information to the next generation by a reproduction operator (crossover), producing offsprings. Each offspring is passed through a mutation process. Mutation randomly changes some of the offspring information with a low probability. This operator provides to the Genetic Algorithm the ability for searching near global optimal solutions and consequently, avoid the problem of getting stuck in a local optimum. More detailed information about GA procedure can be found in [19].

2) *Particle Swarm Optimization (PSO)*: Particle Swarm Optimization was inspired by the social behaviour of a flock of migrating birds trying to reach an unknown destination. Each solution is encoded into a “bird” or in this case a so-called “particle”. PSO does not create new population throughout its process as GA. The same particles evolve in their social behaviour based on their movement towards a destination. Physically, this mimics a flock of birds that communicate together as they fly. Each bird looks ahead in a specific direction, and then communicate with the others. By doing so, they can identify the bird flying towards the best location.

The initial locations of all particles (solutions) are distributed uniformly. Initial population update their position based on their current position and their velocity as.

$$x_i^{t+1} = x_i^t + v_i^{t+1} \quad (12)$$

The process is repeated until the flock reaches a desired stopping condition. The velocity of each bird depends on several factors, such as, its previous velocity, the difference between the current position and the best position the particle has reached and, the difference between the current position and the best position in the current neighbourhood. This is expressed as

$$v_i^{t+1} = wv_i^t + c_1r_1(xBest_i^t - x_i^t) + c_2r_2(gBest_i^t - x_i^t) \quad (13)$$

where $xBest$ is the best particle position and $gBest$ is the best group position. The parameter w takes the value between 0 and 1. Moreover, r_1 and r_2 are two random parameters where $r_1, r_2 \in [0, 1]$. Finally, c_1 and c_2 are two acceleration constants typically taken as $c_1 = c_2 = 2$. The initial velocity of a particle can be taken as zero. It is important to note that the process involves both social interaction and intelligence. Birds can learn from their own experience (local search) and also from the experience of others birds around them (global search). More detailed information about PSO procedure can be found in [20]

IV. RESULTS AND DISCUSSION

In this section the main results of the study are delivered. The parameters used for simulation are presented in Table I.

TABLE I
SYSTEM PARAMETERS

Parameter	Value	Parameter	Value
Semi-angle Half power	60°	Lambertian Order	1
Source power	20W	Area of PD	10 ⁻⁴ m ²
$T_s(\psi_{ij})$	1	FOV	120°
Room dimensions	1.4 × 1.4m	LED height	3m
ρ	0.1	Number of LEDs	4
PD height	1m	x-y Grid step	0.2m
MaxIter	20	$g(\psi_{ij})$	1

Each metaheuristic method will run for a maximum number of iterations given by MaxIter. The maximum number of iterations is set to 20 in order to ensure real time operation of the system. Lower bounded, as well as, lower and upper bounded optimization problems of Eq. 9 and Eq. 10 respectively are solved using both metaheuristic methods, GA and PSO.

In Fig. 4 a 3-D distribution of estimated positions is compared with the real position of the mobile node for a PD height of 1 m. As it can be seen, GA methods achieves better performance when dealing with upper unbounded problem of Eq. 9. Even with a large searching space, GA is capable to provide accurate and precise position estimation using an unbounded optimization problem. On the other hand, PSO is not capable to provide a feasible solution within margins of the room size when a maximum number of 20 iterations is imposed as stopping criteria.

On the other hand when room size constraints are considered (Eq. 10) GA shows also better performance than PSO in terms of accuracy. In Fig. 5 the CDF curves of 3-D positioning errors are shown. As it can be seen the GA method combined with map information overcomes PSO by few centimetres. PSO method used for solving 9 has larger error than GA method, due to this, its CDF curve is out the chart.

In Fig. 6 2-D dispersion of solutions are shown. Since GA and PSO does not ensure a local minima, the method

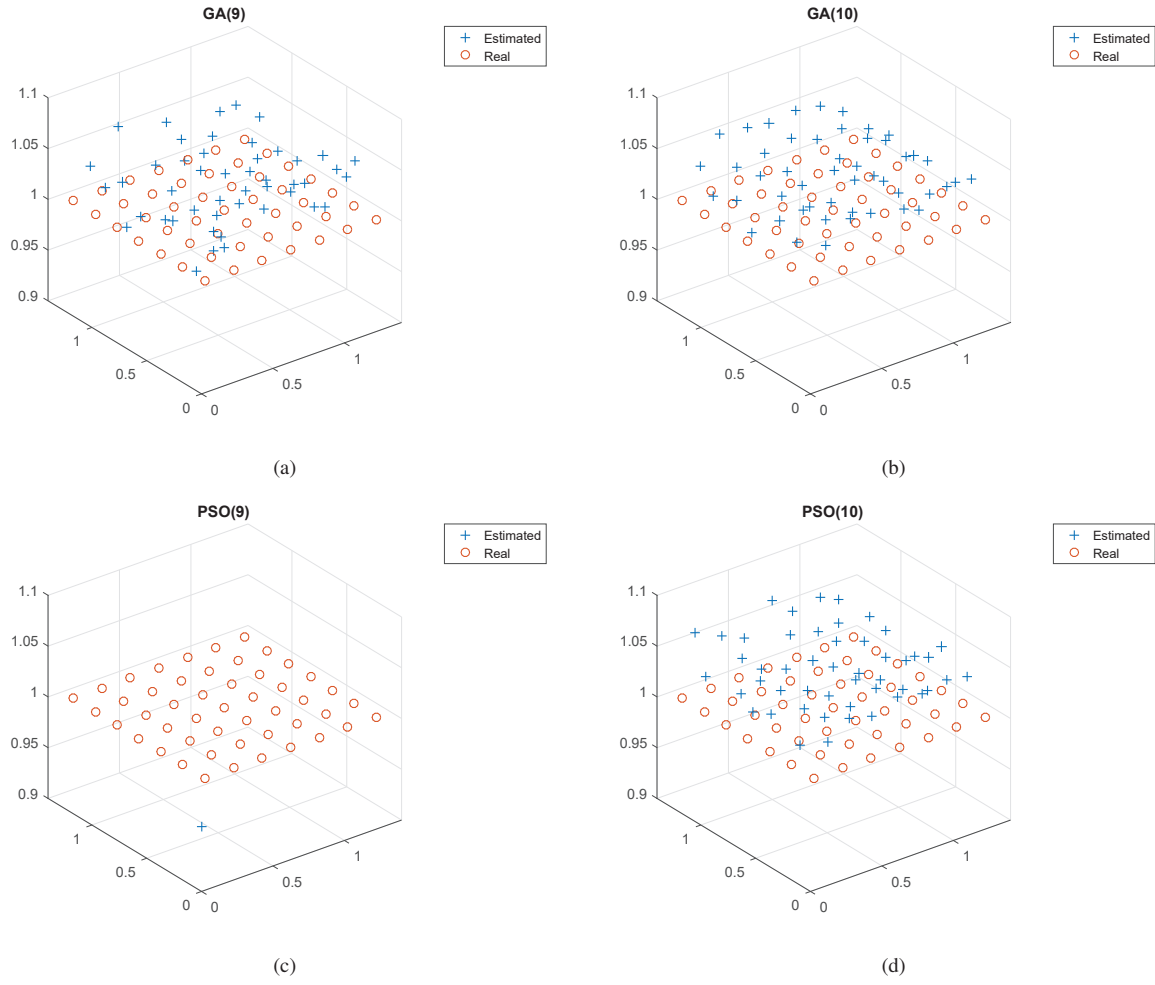


Fig. 4. 3-D distribution of real position and estimated position (a) GA for solving Eq. 9, (b) GA for solving Eq. 10, (c) PSO for solving Eq. 9 and (d) PSO for solving Eq. 10

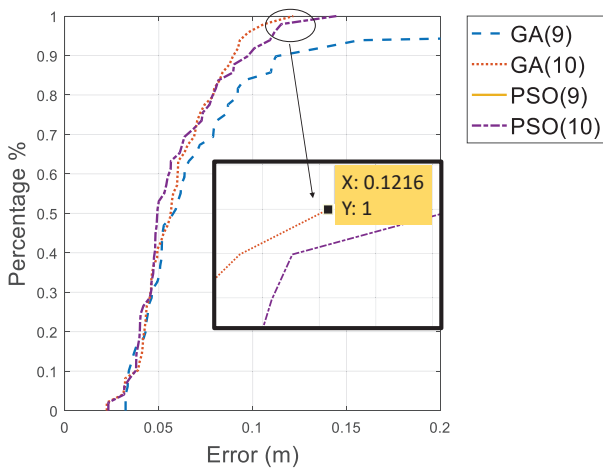


Fig. 5. The CDF curves of 3-D positioning errors (a) GA (b) PSO

performance is highly dependent on initial parameters. In addition to this, the effect of additive noise and wall reflections affect the accuracy of the methods. GA outperforms PSO in terms of precision since, for 100 independent experiments, GA provides more accurate and closer estimates. On the other hand, PSO solutions have higher standard deviation.

V. CONCLUSIONS

In this work a comparison between two classical optimization methods has been presented. Two different optimization problem formulations were used in order to measure its impact in method's performance. GA has shown to be more robust than PSO. They are capable to provide feasible positioning estimates when no map information is used in the optimization problem. On the other hand, PSO is unable to deliver feasible solutions when the optimization problem is not upper bounded. In the upper bounded optimization problem GA finds better solutions than PSO. In addition to this, independent tests have shown that GA solutions have a lower standard deviation than PSO. In future work the proposed methods will be

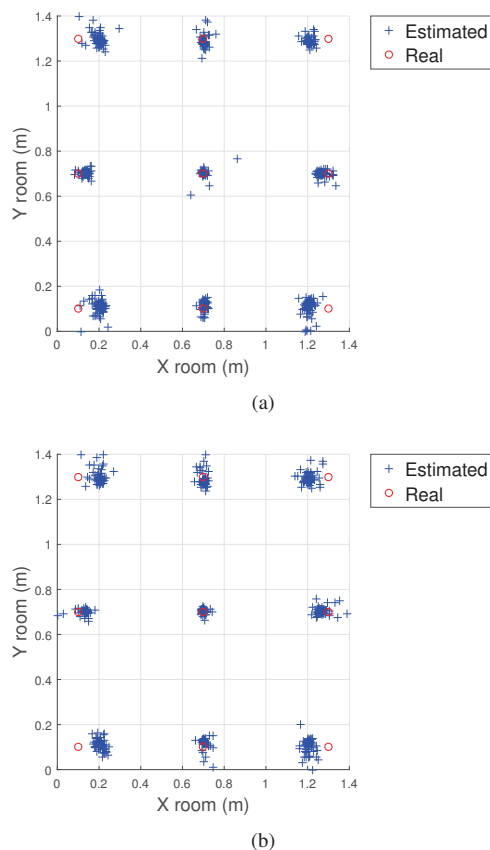


Fig. 6. 2-D dispersion of solutions (a) GA for solving Eq. 10, and (b) PSO for solving Eq. 10

implemented and tested in real scenario. In addition to this, tilting receivers will be considered and the fitness functions will be modified in order to take into account possible effect of this angle in the system performance.

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